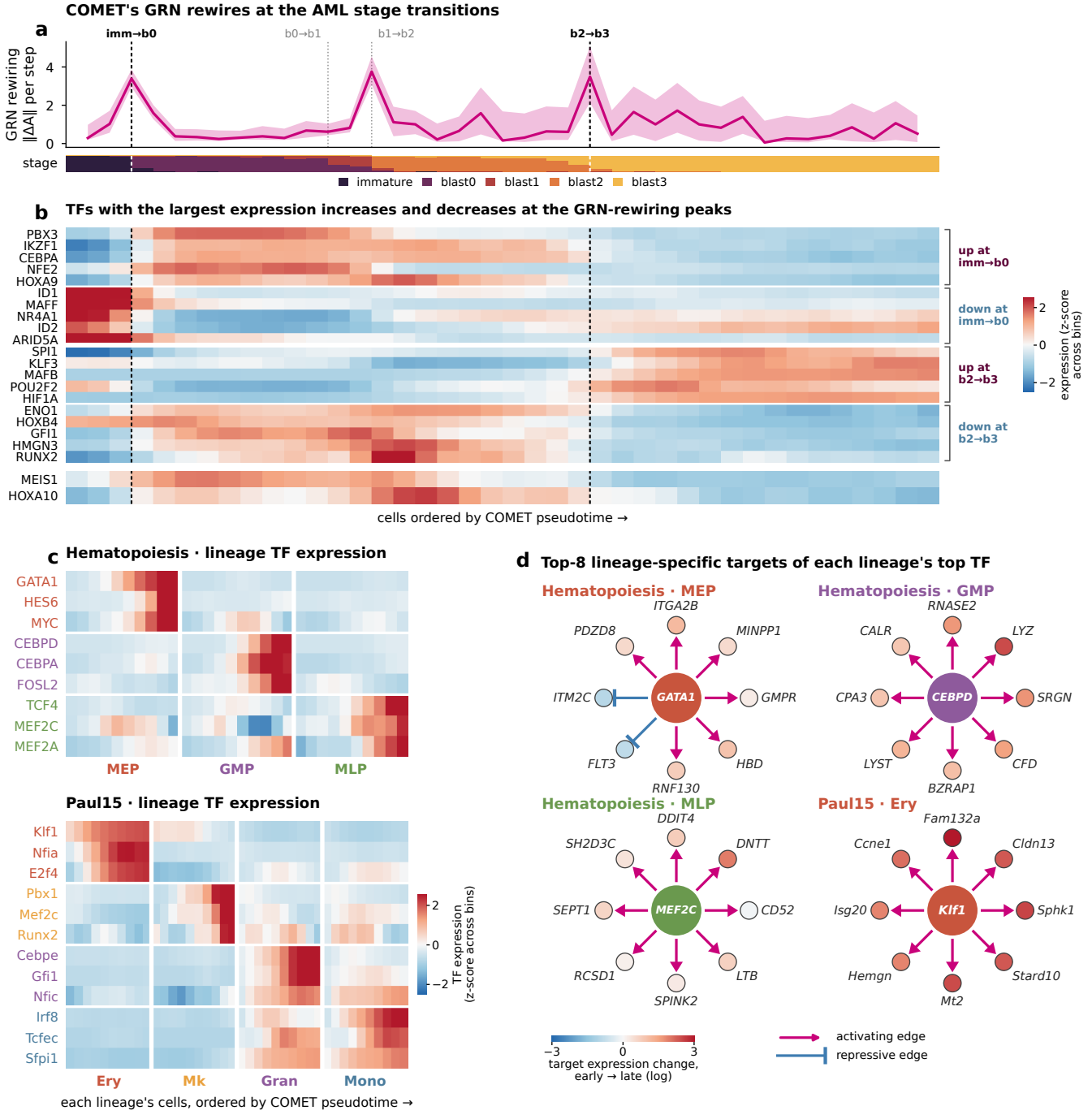

COMET's GRN rewires at the AML stage transitions
a
imm→b0
b0→b1
b1→b2
b2→b3
GRN rewiring
‖ΔA‖ per step
stage
immature
blast0
blast1
blast2
blast3
b
TFs with the largest expression increases and decreases at the GRN-rewiring peaks
PBX3
IKZF1
CEBPA
NFE2
HOXA9
ID1
MAFF
NR4A1
ID2
ARID5A
SPI1
KLF3
MAFB
POU2F2
HIF1A
ENO1
HOXB4
GFI1
HMGN3
RUNX2
MEIS1
HOXA10
up at imm→b0
down at imm→b0
up at b2→b3
down at b2→b3
expression (z-score across bins)
cells ordered by COMET pseudotime →
c
Hematopoiesis · lineage TF expression
GATA1
HES6
MYC
CEBPD
CEBPA
FOSL2
TCF4
MEF2C
MEF2A
MEP
GMP
MLP
Paul15 · lineage TF expression
Klf1
Nfia
E2f4
Pbx1
Mef2c
Runx2
Cebpe
Gfi1
Nfic
Irf8
Tcfec
Sfpi1
Ery
Mk
Gran
Mono
TF expression (z-score across bins)
each lineage's cells, ordered by COMET pseudotime →
d
Top-8 lineage-specific targets of each lineage's top TF
Hematopoiesis · MEP
GATA1
ITGA2B
PDZD8
MINPP1
ITM2C
GMPR
FLT3
HBD
RNF130
Hematopoiesis · GMP
CEBPD
RNASE2
CALR
LYZ
CPA3
SRGN
LYST
CFD
BZRAP1
Hematopoiesis · MLP
MEF2C
DDIT4
SH2D3C
DNTT
SEPT1
CD52
RCSD1
LTB
SPINK2
Paul15 · Ery
Klf1
Fam132a
Ccne1
Cldn13
Isg20
Sphk1
Hemgn
Stard10
Mt2
target expression change, early → late (log)
activating edge
repressive edge